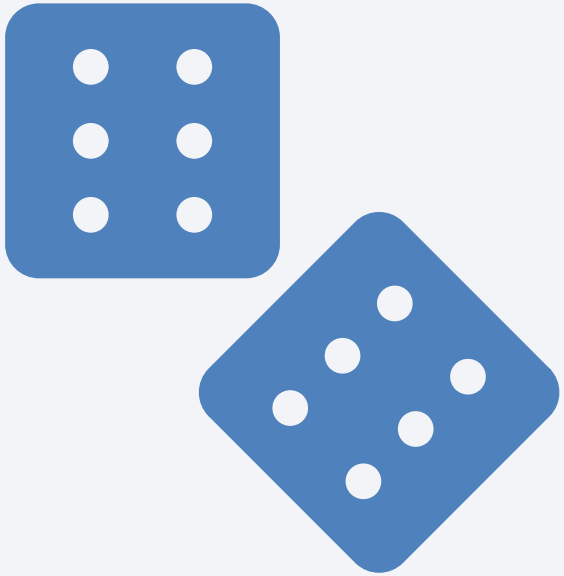
Facebook Live Sellers (Thailand) K-Means Clustering Analysis



Engagement Analysis & Clustering Insights

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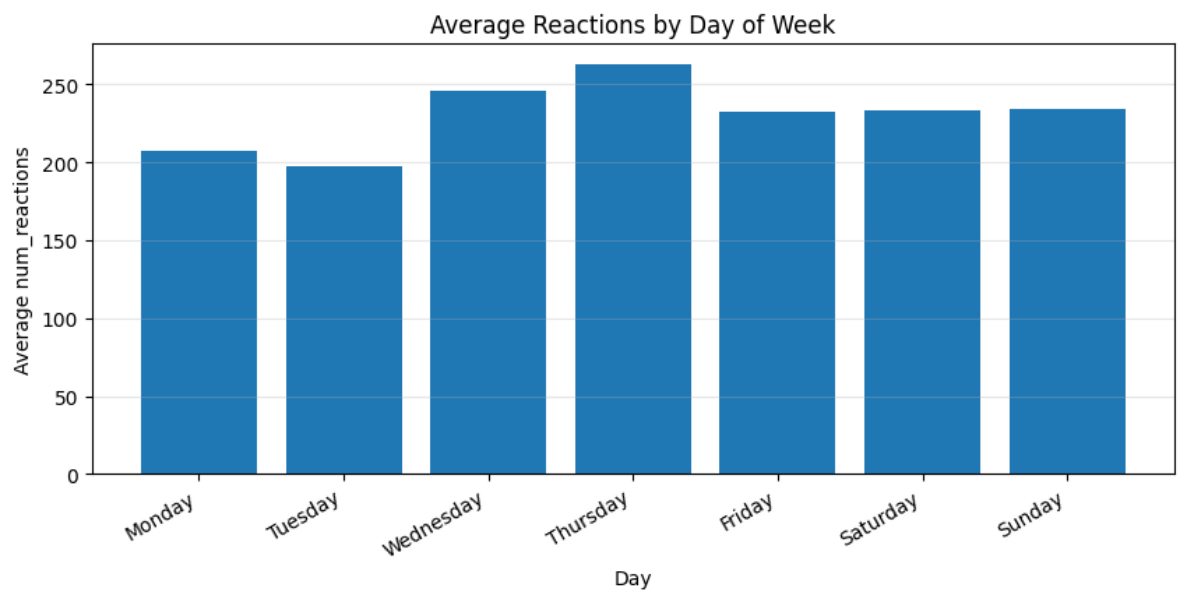
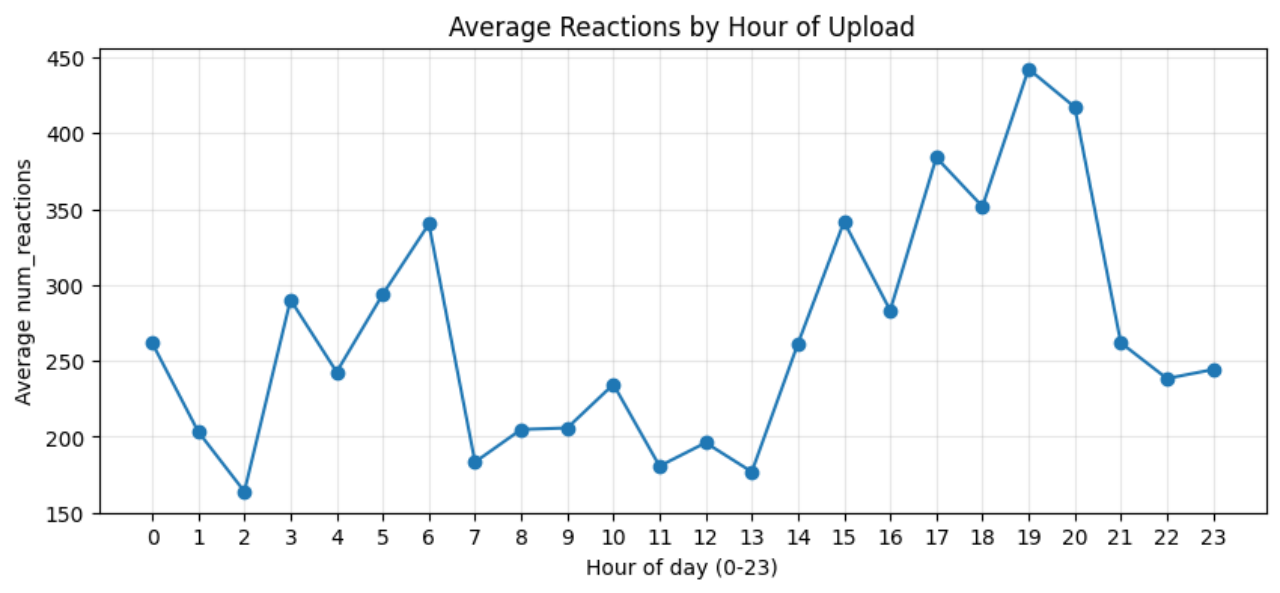
Dataset Overview

- 7050 Facebook posts
- 12 cleaned attributes
- Engagement metrics: reactions, comments, shares
- Post types: link, photo, status, video
- Objective: Analyze engagement & apply clustering

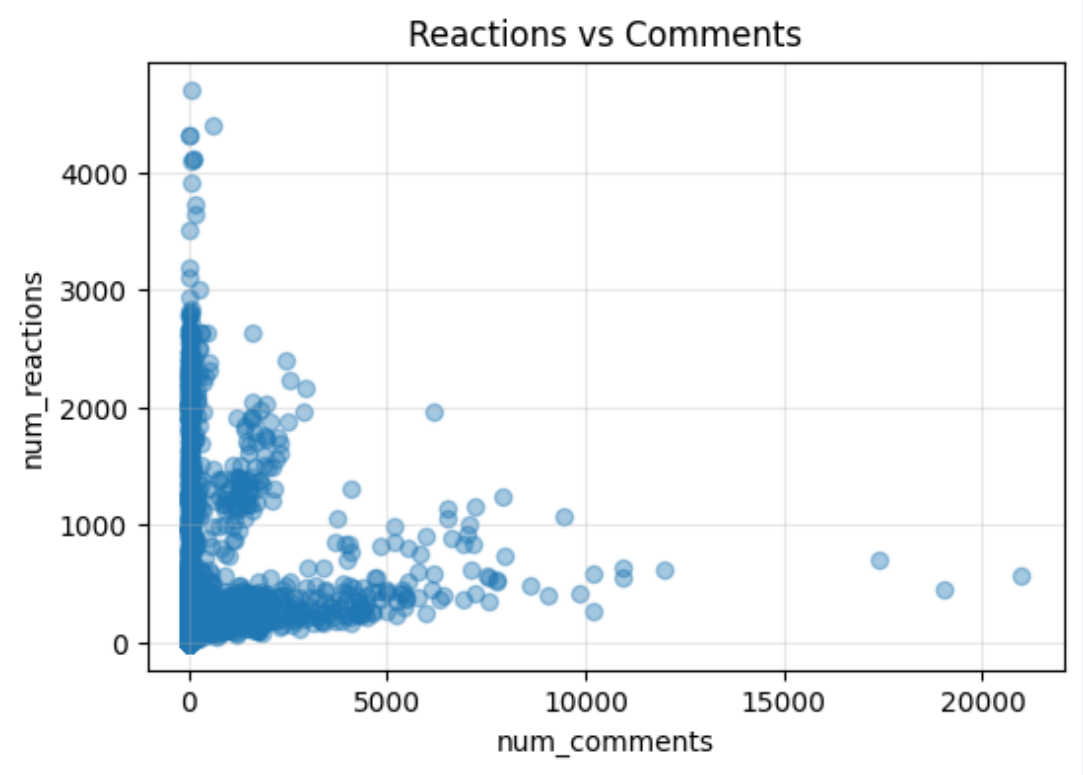
Impact of Upload Time on Reactions



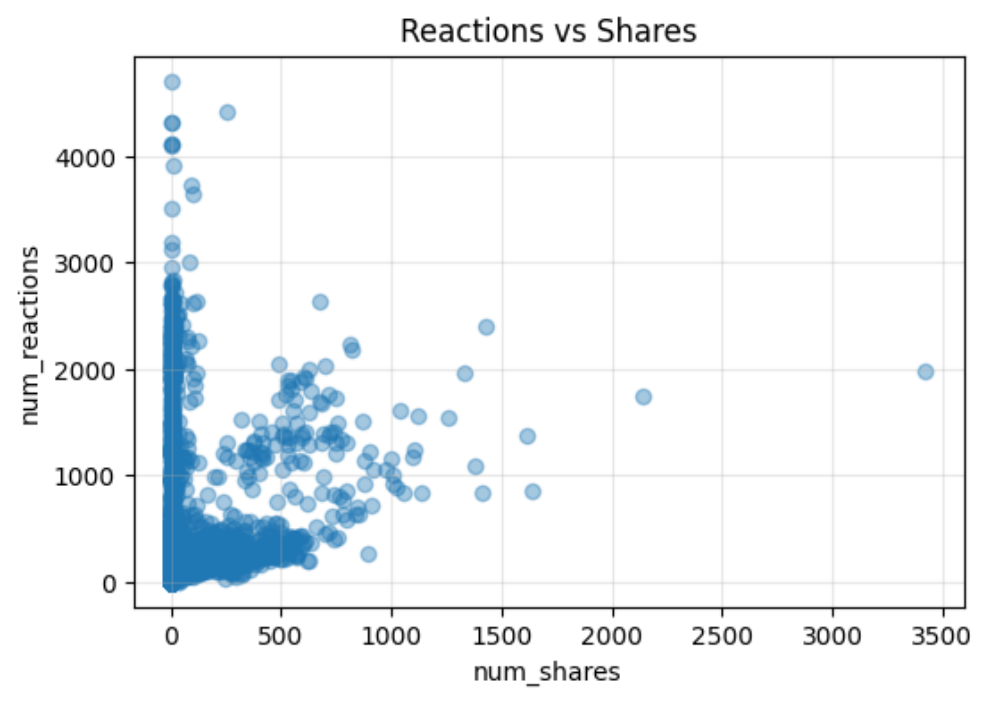


- Reactions vary based on upload hour
- Certain hours show higher average engagement
- Some weekdays perform better than others
- Posting during peak times increases reach
- Timing is critical for engagement strategy

Correlation: Reactions vs Comments & Shares

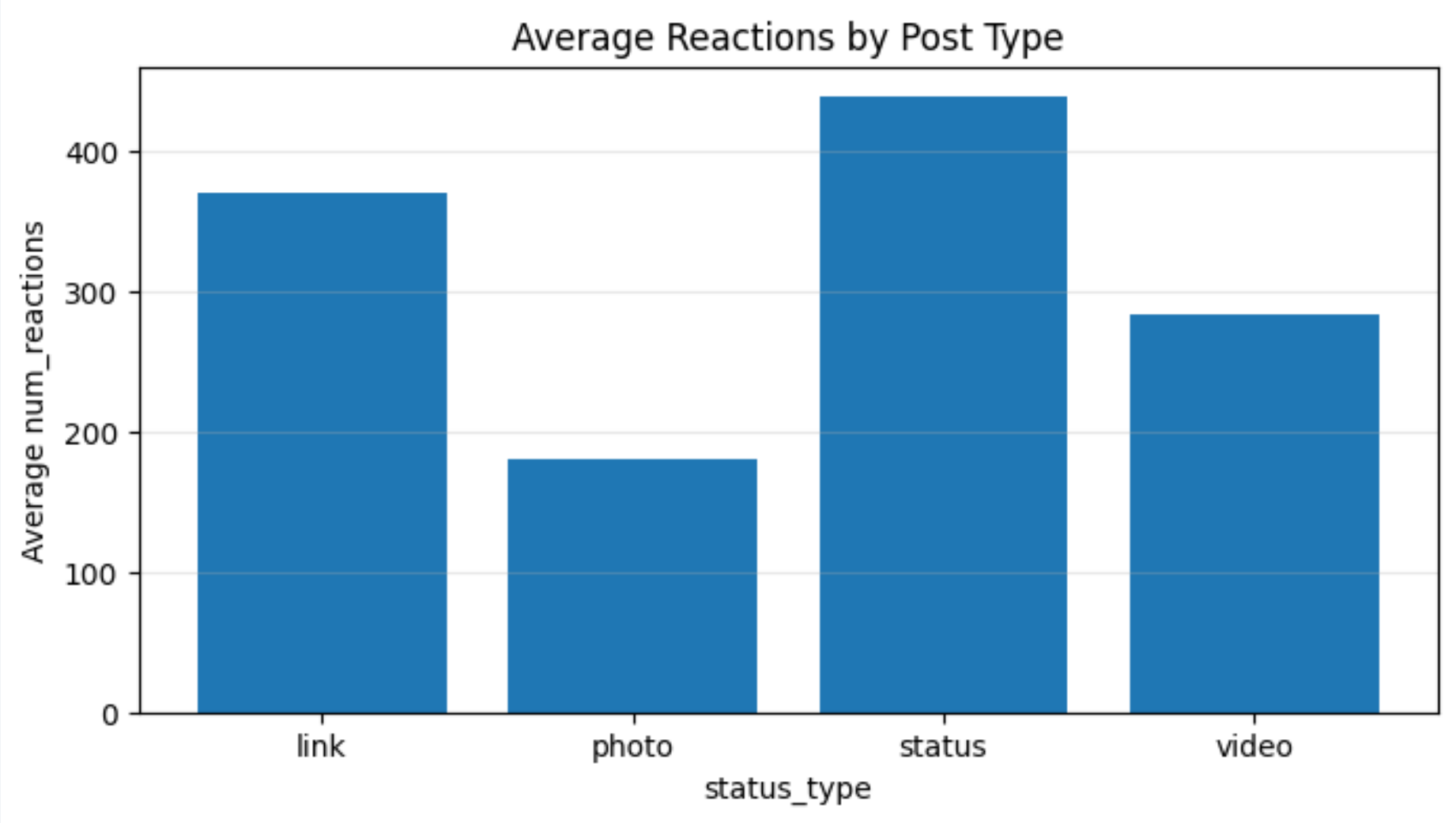


- Engagement metrics are interconnected
- High comments & shares boost reactions
- Viral posts perform well across all metrics

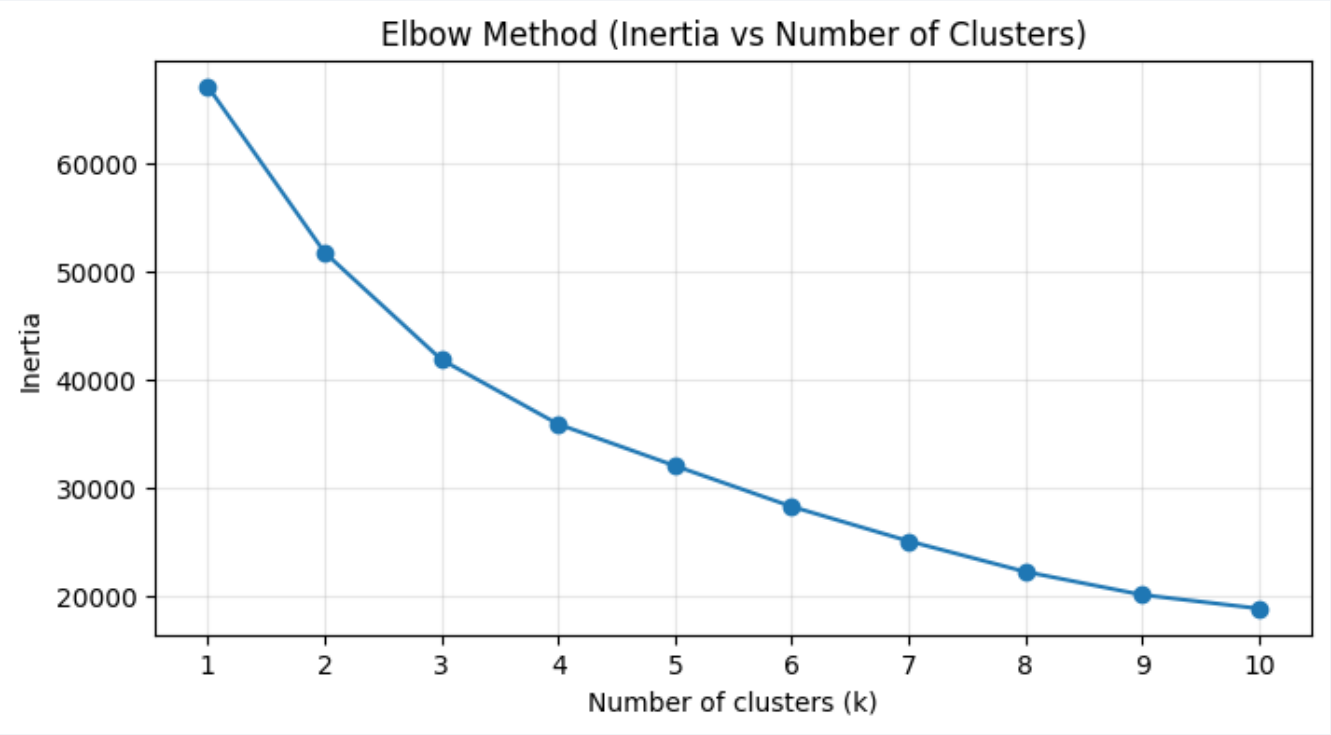


- Strong positive correlation: Reactions & Comments
- Positive correlation: Reactions & Shares

Average Reactions by Post Type



Elbow Method for Optimal Clusters



- Tested cluster values from 1 to 10
- Observed inertia reduction trend
- Identified elbow point
- Selected optimal clusters (k = 3)
- Ensures balanced clustering

K-Means Cluster Insights

- Cluster 1: Low Engagement Posts
  - Low reactions, comments, shares
- Cluster 2: Medium Engagement Posts
  - Moderate interaction levels
- Cluster 3: High Engagement Posts
  - High reactions, strong viral potential
- Clusters help segment content strategy.

Business Recommendations

- Post during peak reaction hours (evening performs well)
- Encourage comments & shares to increase reactions
- Focus on high-engagement post types
- Use clustering for targeted content strategy
- Data-driven scheduling improves ROI